AS2: what E-plus-continuing aggregation can conceal

Elementary VSL lane-drop task review

19 October 2026

Status. AS1 is user-accepted. The bounded AS2 review is completed at its information checkpoint; AS3 has not begun. No traffic experiment, new traffic mechanism or additional observation restriction is introduced.

Core finding. Identical E and combined continuing traffic can conceal different A/B loads even from a complete predecision longitudinal record. A common low-speed preparation can keep early merging donor-limited, so it leaves no trace of unused adjacent receiving. An extra initial adjacent-load statistic removes the ambiguity in this homogeneous family. The result concerns information, not different best advice or useful gradients.

1 Accepted world and the information being compared

Use the user-accepted AS1 initial-family/forward contract. E ends, A is its adjacent continuing lane, B the far continuing lane. At time zero the cohort is uniform on $[0, \ell]$, the terminal and recovery account are empty, and there are no later external arrivals. Put

$$r = e_0 + a_0, \quad \rho = e_0 + a_0 + b_0, \quad M = \ell\rho, \qquad C(v) = \frac{wk_j v}{w+v}, \quad k_c(v) = \frac{wk_j}{w+v}. \tag{1}$$

Keep all AS1 bounds, laws, coefficients and choke-completion TST. The future action is one common constant $v \in [v_-, v_+]$ selected at zero; only information available before that action may choose it.

All predictors receive the same geometry, physical coefficients, action set, initialization-family description, disclosed loading recipe and common past advice. These do not include the realized split between A and B. The nested information roles are:

Role	Available information at the decision
Level 1	Every admitted lane-blind predecision longitudinal density, flow, speed and cumulative-count channel, with all lawful deterministic inference.
Level 2	Level 1 plus initial E mass and combined A+B mass. Any permitted E-resolved history is retained; the construction below even matches that stronger record. The realized A/B split is not supplied separately.
Level 3	The same common record plus individual initial E, A and B masses. No future traffic, queue or recovery trace is supplied.
Compact peer	Level 2 plus one initial adjacent-lane mass or an equivalent scalar. It has the same initial information as Level 3 in this family.

Initial-dependent r is not a shared physical parameter: with known e_0, it discloses a_0. Actual A/B boundary traces, adjacent receiving or a pair label likewise cannot be passed as common inputs in this contrast. If an actually admitted channel reveals the split, its inference must be allowed; it defines an informative case rather than an exception to suppress.

2 The existing loading recipe reaches unequal A/B states

2.1 A general sufficient construction

Select two AS1-admitted initial states with common $e_0 = e > 0$ and $a_{0,i} + b_{0,i} = c$, but $a_{0,1} \neq a_{0,2}$. Thus $\rho = e + c$ is common and $r_i = e + a_{0,i}$ differs. Use the already disclosed predecision H1 loading recipe, with common past speed $\bar{v} > 0$, $T = \ell/\bar{v}$, and an empty road before $-T$:

$$e^{\mathrm{in}}(t) = e\exp(-\bar{v}t/h), \quad a_i^{\mathrm{in}}(t) = r_i - e^{\mathrm{in}}(t), \quad b_i^{\mathrm{in}}(t) = b_{0,i}, \qquad -T \le t \le 0. \tag{2}$$

All inputs cease at zero. These are exogenous preparation traces; they do not depend on which future advice is tested. The recipe is the same as the accepted historical construction, now evaluated on the independent initial family. It adds neither a preparation mechanism nor a new controller.

Sufficient loading conditions are

$$e\exp(\ell/h) < \min_i r_i, \qquad \max_i r_i + (\gamma - 1)e\exp(\ell/h) < k_c(\bar{v}), \qquad b_{0,i} \le k_c(\bar{v}), \quad \ell < L_p. \tag{3}$$

For $\gamma \ge 1$, these ensure positive A input and free lane densities. The donor test for H1 is $a + \gamma e \le k_c(\bar{v})$. It holds strictly because $a_i + \gamma e = r_i + (\gamma - 1)e$.

The occupied front is $x_f(t) = \bar{v}(t + T)$. In $0 < x < x_f(t)$ the homogeneous fields are

$$e(x,t) = e\exp(-\bar{v}t/h), \quad a_i(x,t) = r_i - e(x,t), \quad b_i(x,t) = b_{0,i}. \tag{4}$$

They satisfy the same transport and lateral balances:

$$e_t + \bar{v}e_x = -\bar{v}e/h, \quad (a_i)_t + \bar{v}(a_i)_x = \bar{v}e/h, \quad (b_i)_t + \bar{v}(b_i)_x = 0. \tag{5}$$

For example a characteristic entering at $t - x/\bar{v}$ carries $e^{\mathrm{in}}(t - x/\bar{v})e^{-x/h} = e\exp(-\bar{v}t/h)$. The front travels freely into an empty road. It never reaches L_p before zero, so the terminal/recovery history is empty. At zero the required uniform independent-lane initial state is reached, with total mass M.

2.2 A nonempty example inside the accepted AS1 family

Reuse AS1's shared scales and coefficients:

$$\begin{gathered} v_- = u,\ v_+ = 2u,\ w = 2u,\ k_j = 3\kappa,\ \gamma = 2, \quad R = 39\kappa u/20,\ R_B = \kappa u, \\ L_p = 2\ell,\ b = \ell,\ c_0 = 11/10,\ \beta = 1/100, \quad e = 3\kappa/5, \quad a_{0,i} + b_{0,i} = 11\kappa/20, \\ (a_{0,1}, b_{0,1}) = (\kappa/5, 7\kappa/20), \quad (a_{0,2}, b_{0,2}) = (7\kappa/20, \kappa/5). \end{gathered} \tag{6}$$

Use $\bar{v} = u$ and any $0 < \ell/h < \log(4/3)$. Then $r_1 = 4\kappa/5$, $r_2 = 19\kappa/20$, $e\exp(\ell/h) < 4\kappa/5$ and

$$r_2 + e\exp(\ell/h) < (19/20 + 4/5)\kappa = 7\kappa/4 < k_c(u) = 2\kappa. \tag{7}$$

The entire continuous AS1 split interval between these endpoints satisfies the same loading bounds. All postdecision AS1 admission margins remain unchanged. This is a sufficient matched-history subfamily, not a restriction of the whole admitted task to those histories.

For exact rational checks one may set $\ell/h = \log(6/5)$. Earliest E input is then $18\kappa/25$, the smallest A input is $2\kappa/25 > 0$, and the worst donor work density is $167\kappa/100 < 2\kappa$. At the decision, future advice $2u$ has the same AS1 initial offer $6\kappa u/5$ in E, but receiving alternatives $13\kappa u/10$ and $23\kappa u/20$. Thus identical pasts can precede the already proved distinct initial receiving response. This local calculation is not an action comparison.

3 The complete observation record is identical

Proposition 1. *The constructed pair has identical lane-blind longitudinal fields and counts at every cross-section for every predecision time. It also has identical E and combined A+B histories, even if these richer histories are available. Nevertheless its initial A/B split is different.*

Proof. Write $I(x,t) = \mathbf{1}_{\{0<x<x_f(t)\}}$ for $-T \le t \le 0$. The common aggregate fields and cumulative longitudinal count are

$$\begin{aligned} k_\Sigma(x,t) = \rho I(x,t), \quad Q_\Sigma(x,t) = \bar{v}\rho I(x,t), \\ N_\Sigma(x,t) = \rho[x_f(t) - x]_+. \end{aligned} \tag{8}$$

Speed is $\bar{v}$ on the occupied region; empty-region masks are the same. E fields are $k_E = e(t)I$, $Q_E = \bar{v}e(t)I$. For a reached cross-section, its cumulative E count is

$$N_E(x,t) = \bar{v}\int_{-T+x/\bar{v}}^{t} e(s)\,\mathrm{d}s; \quad N_E = 0 \text{ before its first arrival.} \tag{9}$$

This is the integral of the *local* E flow; it is not a count of conserved E identity through upstream transfers. Since $e(t)$ is common, all E-resolved channels in this expression are common. Continuing fields are $k_{A+B} = (\rho - e(t))I$, $Q_{A+B} = \bar{v}k_{A+B}$ and $N_{A+B} = N_\Sigma - N_E$, likewise common.

The represented early E-to-A transfer density source $\bar{v}e(t)/h$ is also common, so its cumulative transferred mass is not a distinguisher for this pair. This does not claim equality of unmodeled microscopic gap or individual maneuver observations. Before $-T$ both roads are empty; through zero both terminal/recovery histories are empty. At zero, $E_0 = \ell e$ and $M - E_0 = \ell c$ agree, while A/B differ. Any deterministic filter or inference from the complete matched record has identical inputs. The result therefore also holds for any finite sensor/history subset of that record, without shortening the history window. □

The common predecision population-time is $MT/2$, independent of future advice. Excluding it from the decision-time TST does not remove an action-dependent or pair-dependent term. Nothing from the post-action trajectory has been included in the decision input.

What follows. Level 2 cannot identify the correct initial A/B split for every member of this admitted pair. It may retain uncertainty or use a disclosed prior; it is not required to assume equal A/B loads. No deterministic point estimate can be correct in both states from identical inputs. This does not imply different optimal actions: a common action could be optimal in both, and AS3 must check that separately. The construction establishes an exact conditional ambiguity, not universal nonobservability in traffic.

4 The compact peer closes the initial-information gap

Let E_0, A_0, B_0 denote initial masses, so $M = E_0 + A_0 + B_0$. Given Level 2's common M, ℓ, E_0 and one adjacent mass A_0,

$$e_0 = E_0/\ell, \qquad a_0 = A_0/\ell, \qquad b_0 = (M - E_0 - A_0)/\ell. \tag{10}$$

An equivalent scalar such as r also suffices. The AS1 forward law is unique with neutral initial terminal/recovery history. The compact peer and explicit per-lane initialization therefore predict the identical complete trajectory and TST for every admitted advice. Their objective derivatives are equal wherever those derivatives exist; this identity is not a numerical-gradient validation or a new regularity theorem.

Predicted queues, speed marks and retained recovery histories still evolve. Initial scalar sufficiency does not reduce that dynamic state to a single meter reading. Nor does it establish

that full per-lane spatial detail, integer mass, identities or discrete order are necessary for this task. Equal-information representation/optimizer comparisons remain separate from the Level 2-to-Level 3 information comparison.

5 When history does identify A, and the boundary of this result

If an admitted record supplies exact E density and its time derivative while early transfer is *strictly receiving-limited* at known common past speed $\bar{v}$, then

$$\dot{e} = -\frac{C(\bar{v}) - \bar{v}a}{\gamma h}, \qquad a = k_c(\bar{v}) + \frac{\gamma h}{\bar{v}}\dot{e}, \quad r = e + a. \tag{11}$$

The body must be present and this is a density derivative, not the derivative of whole-lane E mass during changing footprint. With the retained free preparation law, r is constant; its earlier inferred value gives current $a_0 = r - e_0$, then $b_0 = \rho - r$. Such permitted inference is fully available to Level 2.

Strict receiving activity is detectable from a transfer rate below the donor offer, $0 < -h\dot{e} < \bar{v}e$. On a donor-only history,

$$\dot{e} = -\bar{v}e/h, \qquad r + (\gamma - 1)e \le k_c(\bar{v}), \tag{12}$$

which bounds r but does not generally determine it. Both constructed members satisfy the bound strictly. At donor/receiver equality the derivative alone does not certify unique receiving activity. With only initial E and no E history, (11) is not an available measurement; the matched-history proof deliberately grants more, rather than relying on that omission. Exact noiseless identification is not a claim of robust estimation from noisy detector data.

Direct A/B counts, a known realized r, an identifying adjacent-receiving measurement or other informative admitted data must also be honored. The pair certificate applies only to the stated matched channels. There is no new observation restriction: the positive result already matches the existing complete lane-blind envelope and a richer E-resolved envelope.

Checkpoint and provenance. The initial-information contract, complete-history existence certificate, conditional inference cases and compact-peer equality complete AS2 at review scope. AS3 must still address complete TST action relevance and the full advice derivative for this asymmetric comparison; no optimal-action split, interior optimum, regret or useful-gradient result is claimed here.

The loading recipe and transport construction reuse Section 2 of the 14 October information/sensitivity note; initialization and complete forward qualification use the accepted AS1 contract. This note's matched-E asymmetric specialization, inference calculation and compact-peer deduction are original model-relative reasoning. No new paper was viewed or new literature model adopted. Existing physical/provenance limitations remain; applied MP6 is unexecuted and SUMO optional. This document stops at AS2's checkpoint, before AS3 or Step 4.